

Promised Delivery and the Price of Slippage

Callsign **VECTOR** · Gator Quant Hacks 2026, Systematic Trading track, sponsored by Webull · 3 October 2026 · github.com/sebastbernal2-ship-it/gqh-delivery-gap
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1. The claim, and who is on the other side

Long-lived infrastructure is sold before it is built, so a schedule is a published promise. When it moves, the cash flow moves: a delayed substation defers a utility's rate base, a slipped turbine defers a developer's revenue, a postponed commencement defers rent. The party holding schedule risk pays the party paid on delivery, and that transfer is what this study measures.

The reason it is not simply priced away is structural rather than clever. The revision hides in a monthly inventory of planned generators and in filing notes; the median revision is **exactly twelve months**, which reads as routine roll-forward rather than as news; the same obligation must be joined across sources by plant and generator identity; and the measured information lag from period end to public disclosure is **33 to 94 days** depending on the field. A reader who waits for a headline never sees the revision at all.

2. What we measured, and it is large

Nine annual vintages of the EIA-860M monthly generator inventory, November 2015 to November 2023, one vintage per year at the same month so comparisons are like for like. Every generator is a row, and a promise that moved is a revision.

Project level, 2015-2023	Value	Meaning
Promises that moved	3,783	3,687 later, 96 earlier
Median slip	12 months	mean 13.2, worst 120
Most common slips	12 / 24 / 6 / 9	797 / 334 / 256 / 177 cases
Left the planned sheet	3,451	mostly by starting to run
Cancelled or postponed	711	formal withdrawals
Promised against realized	64% late	of 721 generators promised in 2015 and running by 2023; median 2 months, worst 83
Concentration	solar 1,682	then onshore wind 435, hydro 371, batteries 362, gas CC 272

A monthly series over 111 vintages adds the second finding: the capacity promised for the next year grew at a **median 41.7 percent per year** across 2016–2024, so slippage never reversed the pipeline. Any naive slippage signal is therefore positive in 89 percent of months, which is the trap in section 4.

The shape of a revision, at monthly resolution. Tracking 6,407 generators across the development vintages gives a sharper picture than the annual comparison can. A first revision arrives quickly, a median of **5 months** after a project first appears, and it is usually small: **36 percent of first revisions are a nudge of one month or less**, while 27 percent are six months or more. Excluding nudges, **44 percent of tracked generators** still had a real revision. The hazard rises with age rather than falling (23 percent in the first six months, 44 percent between one and two years), so a promise is not safer for having survived. Between annual snapshots, where those increments accumulate, the modal move is twelve months.

That decomposition matters for the search rather than only for the description. ****A revision signal built from all revisions is dominated by month-sized noise****, which is one plausible reason the firm-level tests found so little: the median revision carries almost no information, and whatever is there lives in the tail. Any future signal should be defined on revisions of six months or more, and that is a change of definition made on mechanism evidence rather than on a result.

3. Firm level: what a company discloses about itself

Project revisions rarely belong to a listed company: an EIA entity is a developer or operator, and only **6.2 percent** of slipped capacity sits with an entity whose name carries a large listed owner, against **80 percent** held by project and holding companies. So the tradable trace must be what a firm says about itself. SEC XBRL gives that, because every fact names the filing that reported it, which joins to an acceptance timestamp.

Firm level	Filings	Lag to public
PWR remaining performance obligations	45 facts	median 56 days
ETN remaining performance obligations	15 facts	median 33 days
PWR unapproved change orders	26 facts	median 94 days

Remaining obligations fall as work is delivered, so a fall is a revision to the disclosed book, not a delay, and each change is measured against the firm's own typical change.

4. Three tests, and the honest answer each gave

Test one: the disclosing firm's own price

84 timestamped revisions across PWR and ETN, signal lagged to the session after availability, adjusted against a sector benchmark. Negative surprises, the direction the mechanism predicts, show no consistent move at any horizon (25 events, every $|t|$ below 1.3). Positive surprises drift 3.83 percent over twenty sessions, but 70 of the 84 events are one firm across 41 weeks, so the simple explanation is that firm's own multi-year run.

Test two: breadth, by a rule declared in advance

Every filer reporting remaining performance obligations at least three times between 2015Q1 and 2024Q3, grouped by its own SIC code. 1,285 filers qualify, 382 fall in a declared industry group and 903 form a placebo. That yields **5,663 timestamped revisions**, 4,634 usable, 267 sealed and excluded.

Group	Down surprise, 10d	Down surprise, 20d	n
Contractors	—	+2.40%	27
Equipment	+0.82%	+0.30%	300
Electric services	−1.18%	−1.29%	87
Data centre and real estate	+0.87%	+1.84%	996

Fifty cells were examined, being four groups, two directions and five horizons. Nothing is both meaningful and strong in the predicted direction. **A downward surprise, which should be the bad news, is flat or mildly positive in every group.**

Placebo control. Re-running the identical pipeline with each event's date swapped for another date the same firm also published something gives **seven** cells with $|t|$ above 2, against **two** in the real data, and the same average cell mean (+0.284% against +0.281%). The pipeline does not manufacture associations; the real cells are noise at this sample size.

Test three: the aggregate revision, as a relative-value pair

For each month-end vintage, the capacity promised for the next calendar year against the same promise a year earlier. Long contractors and electrical equipment, short merchant generators and large utilities when the promise grows, and the reverse when it slips. Both legs are liquid large caps, dollar neutral.

Signal definition	n	Gross	Net, 10bp	Net, 20bp	Annualised	Sharpe	Max drawdown	Hit
Next year	74	+0.29%	+0.07%	−0.15%	+0.8%	0.04	−37.8%	50%
Current and next year	74	−0.08%	−0.30%	−0.53%	−3.6%	−0.18	−40.0%	47%
Three years	74	+0.13%	−0.10%	−0.32%	−1.2%	−0.06	−43.0%	47%

Diagnostic that matters more than the table. The next-year signal is positive in 89 percent of months, so that row is a nearly static pair, not a test of a signal: both legs returned about +1.9 and +1.6 percent a month and the spread between them nets to nothing. The balanced definition, positive in 50 percent of months, is negative after costs. A selectivity plateau was also run: trading only the largest third of revisions gives +0.56 percent a month net, against -0.15 percent at the median cut, but that is monotone in selectivity with 25 observations and a Sharpe of 0.28, which points at selection rather than at a plateau.

5. The scan: coverage first, then associations

Before measuring anything, the node space is declared: **22 nodes, 231 pairs, of which 116 are measurable and 115 are blocked with a written reason.** A node needs a representation with a clock and an availability rule, or it must explain its absence. The blocked record is the honest map of what this data cannot answer:

Missing family	Why it cannot be measured here
Compute rental price	only current values are public; the history is a paid tier, and the two open datasets that looked promising are empty shells
Spot power price	the market operator did not respond from this host
Interconnection queues	no queue source reached
Positioning and crowding	the only non-crypto perpetual markets reach back thirteen months
Option implied moves	no entitlement for historical chains

In the mechanism window the scan measured **51 series-level pairs**, each against its own generated null: a block shuffle that keeps short-run autocorrelation, and a twelve-month shift that keeps seasonality. Every survivor inside one family is asset co-movement rather than a relation, so the count that matters is the cross-family one, and there it found **none against 0.9 expected.** The two corrections below are what make that statement possible:

- **One series per node.** Three representations of the promise node are highly correlated, so that node could vote three times in one direction. Keeping one representation leaves **zero cross-family survivors** in this window.
- **Specificity.** A relation that appears in unrelated industries is a common factor, not the mechanism. The only mechanism-relevant survivor, firm disclosure revisions against gas prices, was recomputed per industry group: **+0.31 in the data-centre group, and -0.06 in the contractor and equipment groups.** It belongs to the comparison group, not to the mechanism. The other two survivors are the known relations of utilities against rates and copper against rates.

The scan finds real structure in asset co-movement and in known macro links, and **no survivor that belongs to this mechanism.** That is the fourth negative, and the first that searched the space rather than one hand-picked path.

Two studies, two windows, one firewall. The mechanism and strategy study keeps its history from July 2015 and its twenty-four month holdout from October 2022. The compute era study has its own window: history from June 2022, because that is where the compute price archive begins, development to March 2024, and a six month holdout. Re-running everything inside the compute window was measured and rejected, because with a twelve month lookback on twenty-two months of history the strategy has about ten observations, which is not a tested strategy but an untestable one. The rule between them is a firewall: **nothing found in the compute era may change the strategy's design,** because the two windows overlap and a discovery in one would otherwise spend the other's held-out data.

Inside the compute-era window the same scan measured **62 series-level pairs.** Six beat their own null against 3.1 expected; after the canonical and family corrections, **three cross-family survivors remain against 1.1 expected.** Two are the known relations of power equities and copper against rates. The third links firm disclosure revisions to the promised pipeline, which is the mechanism itself, and it is the

only survivor worth a mechanism conversation. It is not evidence: twenty-two months, one surviving relation, and its seasonal control only partly explains it. **The compute price series, the node this window exists for, is now in the study:** a monthly median of provider spot prices per instance family, rebased across families, twenty-two development months, read straight from the archive rather than from a summary. Measured against eleven partners it clears its own null on none of them. Its strongest association is the scarcity basket at the 88th percentile, with a change correlation of -0.36 and a level correlation of -0.49 ; fuel, copper, uranium, rates, both equity baskets, the promise revisions and firm disclosures all sit between the 7th and 76th percentile. Either the relation is absent at monthly frequency, or twenty-two months is too short, or a provider spot price is too noisy a proxy for what compute actually rents for. Naming all three is the honest end of this line, and choosing one without a test would be the mistake this framework exists to prevent.

6. Risk management

- **Position limits, declared in advance and enforced in the chain log:** 2 percent of NAV per core edge, 1 percent for the expression, and a cap of three unmeasured pilot edges at a time.
- **Structure:** dollar-neutral pairs, so market and sector beta are removed by construction rather than asserted. The one thing the pair cannot remove is the shared AI-infrastructure factor, which both legs carry; that is the main tail risk and it is why the legs are chosen from different positions in the value chain.
- **Observed drawdown:** -37.8 percent on the next-year variant and -43.0 percent on the three-year variant, both worse than the return justifies.
- **De-risking rule:** a position closes on the next vintage if the promise it rests on is not reaffirmed, and the whole pair is stood down when the revision series has been one-signed for six months, because that is when the trade is a static bet rather than a signal.
- **Falsifiers, recorded per edge before measurement:** if the revision stops predicting the firm's own reported obligations, or if the price move is explained by market and sector movement after benchmarking, the study reports a negative rather than a weaker version of the same claim.

7. Liquidity and capital

Both legs are large-capitalisation US listings with deep daylight liquidity: contractors and electrical equipment on one side (PWR, EME, MTZ, DY, ETN, J, HUBB, ROK), merchant generation and utilities on the other (NRG, VST, D, SO, NEE, EXC). Monthly rebalancing with a position change in roughly one month in seven gives about 1.1 times monthly turnover, so costs are small in absolute terms: 10 basis points per leg per rebalance costs about 22 basis points a month in the worst case where both legs flip, and the tables above report the same run at double that. The binding constraint on capacity is not liquidity, it is the absence of an edge: a pair trading for years at a 0.04 Sharpe cannot carry meaningful capital, and the honest statement is that capacity is moot until an association survives its own placebo.

8. Evidence discipline

- **Variants tried, stated in one place:** docs/variants.md, generated from the artifacts. Across every search, **135 comparisons produced 6 survivors against 5.5 expected by chance**. The control is reported separately because it measures the pipeline rather than the search, and it produced **7 against 2.5 expected**, which says our calibrated threshold is on the loose side for autocorrelated series and is why the family correction and the specificity test are not optional. Measurements with no null test are listed apart and excluded from those counts.
- **Every signal lagged:** a filing is usable at its acceptance timestamp plus a declared processing lag; a monthly inventory is usable at the end of its own month. Fills are the close of the first session strictly after that moment, and the code refuses same-bar fills.
- **Point-in-time throughout:** vintages are read as published, never as revised series.
- **Out-of-sample, defined and still closed:** the sealed window is the shorter of a fifth of the series or two years, which here is the final 24 monthly vintages, 2022-10 to 2024-09. Nothing in this note uses them. One command opens it, once, and reports whatever it says:

```
python scripts/run_capacity_strategy.py --open-sealed
```

- **Reproducible:** twelve test suites over the same code paths; the data are public, with a source receipt on every row.

9. What this study concludes, and what it does not

The phenomenon is real, large and measurable: promises move, they move in whole-year steps, and two thirds of promised generators arrive late. Four increasingly broad tests then fail to turn that into a return signal, the last of them searching the declared node space rather than one hand-picked path, and a placebo shows the failures are not an artefact of the pipeline. The most useful sentence this study can offer a reader is therefore a measurement rather than a strategy: **the slippage in this buildout is too slow and too routine to trade against the firms that disclose it, but it is large enough to change what should be built and where.**

Design-time exposure, disclosed. The aggregate capacity signal was defined while the full series was visible, including the months that later became the holdout, so the holdout is clean for **evaluation** and not for **definition**. The evaluation itself used development months only, and the definition is frozen. A fresh holdout would be needed before any claim about that signal could be called out-of-sample in the strict sense.

Limits, stated plainly. Attribution stops at the party that reports the obligation, which is why the project-level route was abandoned at a 6.2 percent coverage ceiling. Obligations fall when work is delivered, so a revision is not a delay. The sealed window is 24 months, which is small. The study has no intraday or option data, because no entitlement was available, so horizons inside a session were never tested. And the modal one-year revision is a descriptive fact; its cause, whether reporting convention or real slippage, remains open.

Appendix: data, code and reproduction

Input	Role	Receipt
EIA-860M monthly generator inventory, 111 vintages	the promise and its revision	eia.gov/electricity/data/eia860m/
SEC EDGAR submissions and XBRL company facts	firm disclosures, with acceptance times	data.sec.gov
Daily adjusted closes	price side, sector benchmarks	provider daily bars

```
make check          # gates: secrets, ledger, structure, paths, owners, chain, link
make test           # twelve suites
python scripts/build_capacity_expectations.py    # 111 vintages to one aggregate panel
python scripts/build_delivery_panel.py --years 2015-2023
python scripts/build_rpo_universe.py && python scripts/build_rpo_events.py
python scripts/run_group_event_study.py --placebo
python scripts/run_capacity_strategy.py
```

Negative results are recorded in docs/decisions.md with their numbers, and the claim path with a falsifier per edge is in docs/chains/t-capacity-revision.jsonl. A submitted PDF is generated from this file; the repository is the deliverable of record.